Always short a 25-delta put, ignoring the VRP signal

Experiment report · L/S Gamma Desk, QUANTT

2026-10-17

Question: On the 25-delta short-put book with the 10-share hedge band, does being short every day regardless of the VRP signal raise net P&L over the signal-timed book, at 1x and 2x spreads, in both halves, without a worse daily tail?

Method:

- **Pre-registration:** the hypothesis card was committed before any run (commit 4e3e599). No engine code was added.
- **Data:** the paper rules (`configs/paper.toml`), with the changes below, are replayed on R2 replay snapshots v1 (config_hash b040760187f0) over the development window 2014-06-02 to 2024-12-31. The hold-out (2025-01-01 to 2026-07-10) was not used.
- **Book:** short-only, a 25-delta short put (`[backtest] put_delta = 0.25`), the 10-share hedge band, and hedges once a day at 10:15 ET.
- **The change:** `[signal] short_band = -inf`. The signal is always short, the flat exit cannot fire, and the book re-enters on the same run after every exit, so it rolls continuously. The 7-DTE time exit, the 1x-credit stop, the 50% profit-take and the 1-day cool-off are the live values, unchanged; the backtest's pre-hold-out embargo also applies.
- **Variants:** `always25` (primary), and `always_atm` (sensitivity, reported only: the same rules with the ATM put).
- **Reference:** the registered `2026-10-17-otm-put/put25` runs (signal-timed 25-delta put), reused, not rerun. Baseline trial 1 is shown for context.
- **Runs:** every variant ran once at 1x and 2x spreads. Costs are $0.65 per contract and $0.005 per share, with a $1 minimum per order.

Correction (2026-10-18): the first version compared the two books day by day with each day's P&L credited to the wrong side of that day's decision. This put the move that triggers a short signal (an IV jump) inside the timed book's holding period. It led to the wrong conclusion that the signal's short days were bad days to be short. The Analysis below now credits P&L to the interval after each 10:15 decision, and that conclusion is withdrawn. The pass bar, the verdict and every number in the tables are unchanged, because full-window P&L and ES over all days don't depend on how days are lined up.

1 Result vs the pass bar

`always25` against `put25` at the same spread multiple. P&L is net USD on daily equity for 1 contract. ES is the mean of the worst 5% of daily P&L, over all days.

Check	1x value	1x bar	2x value	2x bar	Result
Net P&L, full window	+1,316	$> +180$	+862	> -273	pass
Net P&L, 2014-06 to 2019-12	+553	> -280	+371	> -410	pass
Net P&L, 2020-01 to 2024-12	+763	$> +460$	+491	$> +137$	pass
abs ES 5% $\leq 1.10 \times$ reference	121.2	≤ 79.2	123.8	≤ 80.5	**fail**
Event floor: round trips (1x)	187	≥ 30			met

Verdict: failed, on the tail guard only: the daily ES is 1.68x the reference's at 1x and 1.69x at 2x. Every P&L check passes, in both halves and at both spreads. `always25` is the first book in the desk's backtests with positive net P&L at 2x spreads (+$862). The PM chose before the run to keep the ES limit at 1.10x.

1.1 Full-window numbers (not part of the bar)

	put25 1x	**always25 1x**	**always_atm 1x**	**put25 2x**	**always25 2x**	**always_atm 2x**
Net P&L	+180	**+1,316**	−845	−273	**+862**	−3,812
Sharpe on $250k	+0.04	+0.19	−0.10	−0.06	+0.12	−0.45
Max drawdown	1,237	2,041	3,589	1,431	2,195	5,280
ES 5% (daily, all days)	−72.0	−121.2	−142.1	−73.2	−123.8	−157.7
Days holding a position	27%	99%	99%			
Round trips	105	187	178	105	187	178
Hedge orders	340	609	850	340	607	848

1.2 P&L by window (not part of the bar)

Window	**put25 1x**	**always25 1x**	**always_atm 1x**	**put25 2x**	**always25 2x**	**always_atm 2x**
2014-06 to 2015-05	+85	+413	+256	+77	+386	+102
2015-06 to 2021-12 (archive window)	+761	+1,361	+1,707	+428	+1,022	+227
2022	−1,039	+16	−1,576	−1,123	−211	−2,387
2023-2024	+374	−475	−1,231	+345	−335	−1,754

2 Charts

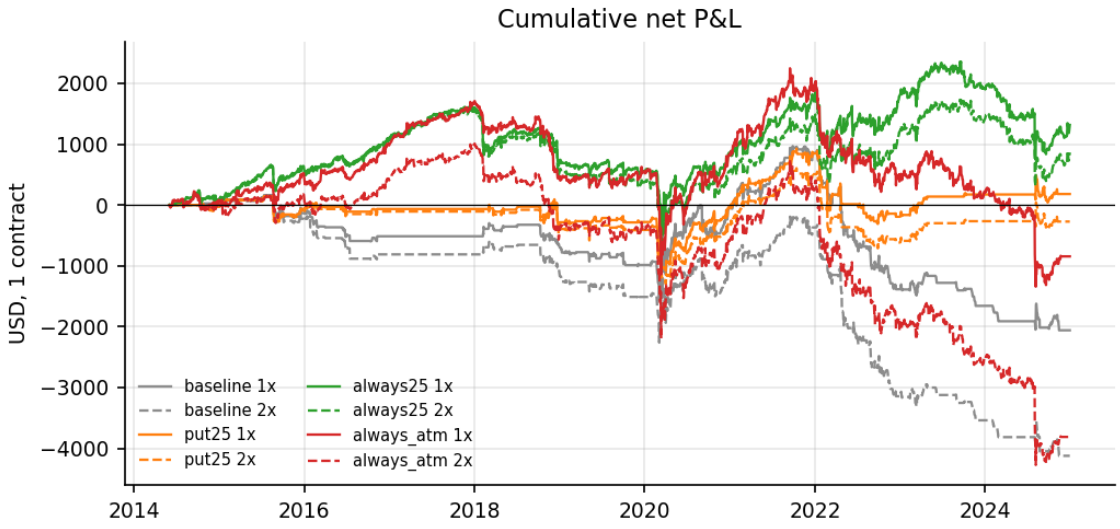


Figure 1: Cumulative net P&L, 1x solid and 2x dashed; baseline (live rules) grey

`always25` (green) rises steadily from 2014 to early 2018, where the timed books are flat or falling. It ends 2017 at +$1,597 at 1x, against −$66 for the timed book. It gives part of that back in 2018 and the March 2020 crash, then peaks at +$2,355 in September 2023. The ATM version (red) tracks it until 2021, then falls hard in 2022 and 2024.

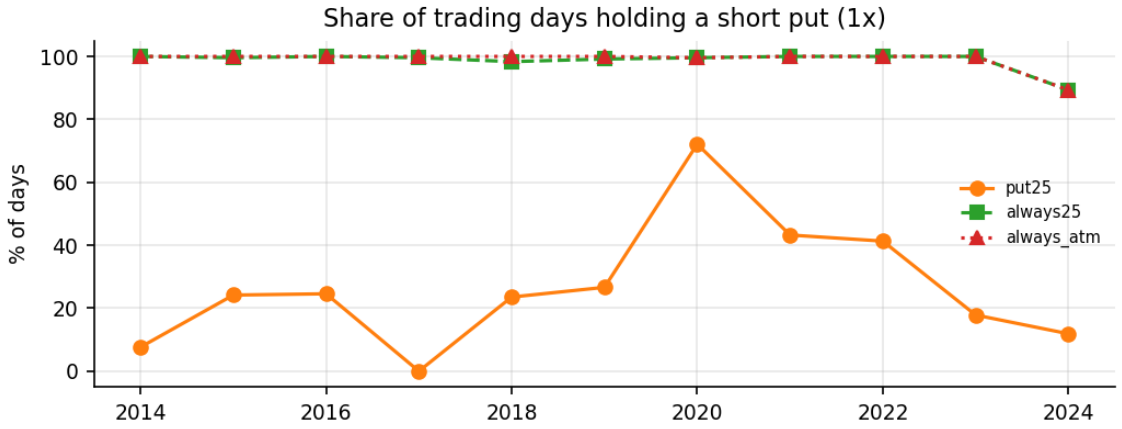


Figure 2: Share of trading days holding a short put

This chart is the experiment. The timed book holds a position on 27% of days, from 0% in 2017 to 72% in 2020. The always-short books hold one on 99% of days. The dip to 89% in 2024 is the embargo: no entries in the last 45 days before the hold-out.



Figure 3: Net P&L by window

Always-short wins in every window except 2023-2024. In 2015-2021, the archive's window, it makes +$1,361 at 1x. In 2022 it roughly breaks even (+$16 at 1x), where the timed book lost $1,039: the timed book was short in the worst stretches of the sell-off, while the rolling book also collected premium on the days in between. 2023-2024 loses $475 at 1x, mostly in the August 2024 spike.

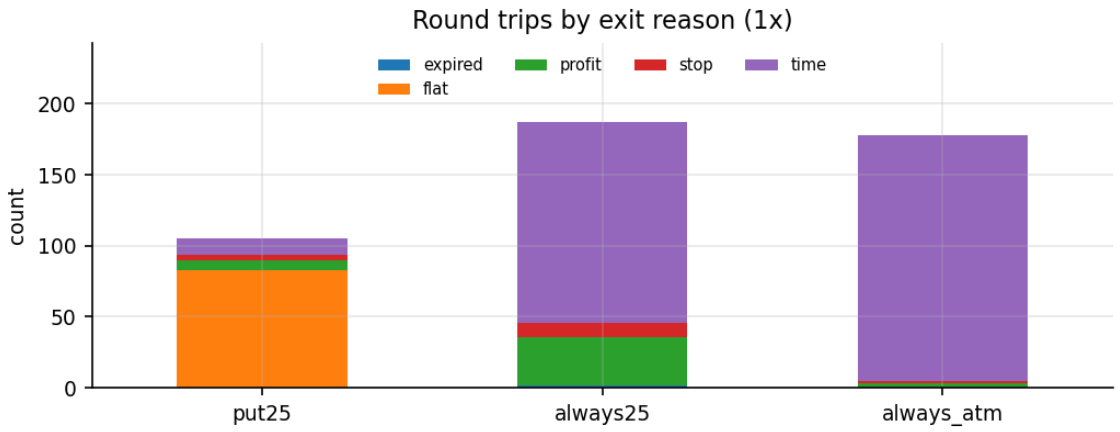


Figure 4: Round trips by exit reason

The rolling book closes 141 trades at the time exit, takes profit 34 times (+$3,710), is stopped 10 times (−$3,119) and holds 2 to expiry. The timed book closed most of its trades on the flat signal.

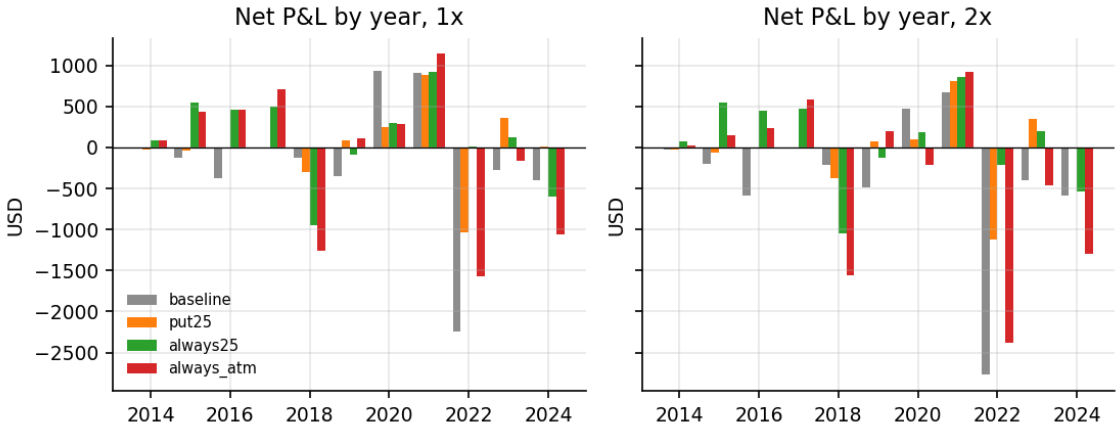


Figure 5: Net P&L by year

At 1x `always25` makes money in 2014-2017 (+$1,597 together, against −$66 for the timed book), 2020, 2021, 2022 and 2023. It loses in 2018 (−$953, in February and December), 2019 (−$91) and 2024 (−$597).

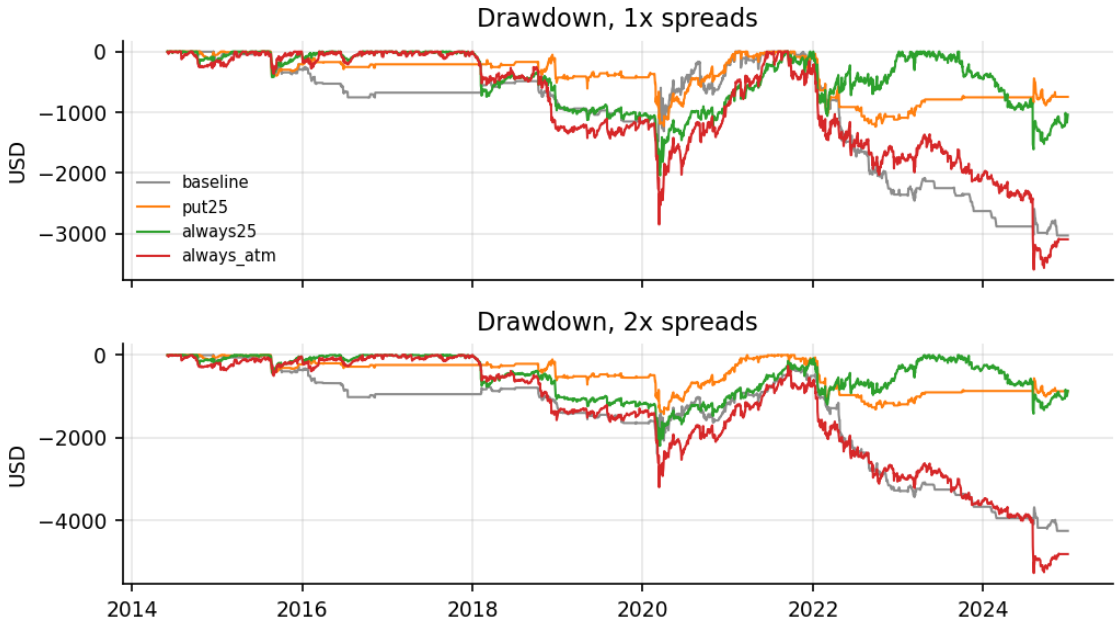


Figure 6: Drawdown at 1x and 2x spreads

The maximum drawdown is larger: $2,041 against $1,237 at 1x, with the trough at the March 2020 crash (2020-03-16). It is still smaller than the live baseline's $3,027.

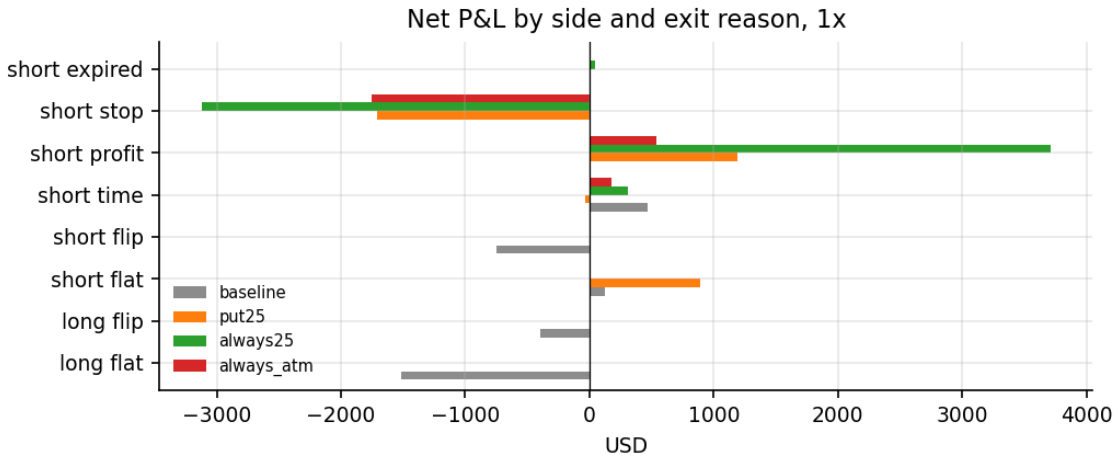


Figure 7: Net P&L by side and exit reason, 1x

Profit-takes and stops net +$591, more than the time exits (+$310 net). Round trips add up to $946 of the $1,316 equity P&L at 1x. The other $370 sits outside any round trip, most likely shares left inside the 10-share band and the hedge unwinds.

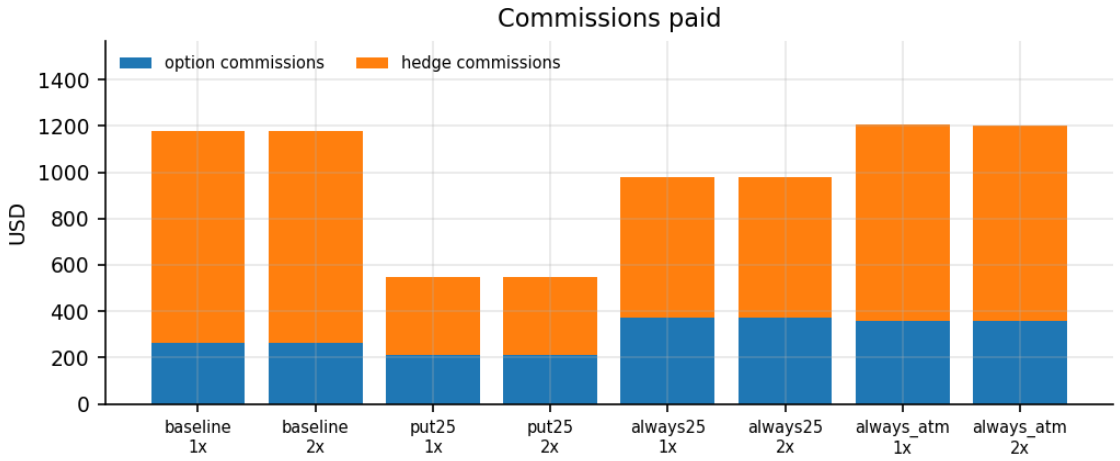


Figure 8: Commissions paid

Commissions rise from $550 to $981, because the book trades about 1.8 times as many round trips and hedges almost every week.

3 Analysis

- **Sitting out cost more than it saved.** Being short every day beats being short only when the signal says so, on every P&L check, at both spreads. Before costs the rolling book earns about +$2,750 against +$1,183. The gap comes from the years the signal mostly sat out: in 2014-2017 the rolling book made +$1,597 against −$66, which is more than the whole net gap (+$1,136). From 2018 to 2024 the rolling book is $527 behind. The archived v1 backtest pointed the same way, but it inspired this test, so the agreement is weak evidence.

- **The bigger tail is real, not just more exposure.** Daily ES over all days is −$121 against −$72, and part of that is the timed book being flat 73% of the time. So compare the two books over the same 798 intervals in which the timed book held a position, each interval running from one 10:15 run to the next. There the rolling book's ES is −$158 against −$135, and it lost $431 where the timed book made $319. The timed book's other −$139 is the cost of its 87 entries from flat (its other 18 entries were re-entries while already holding). The rolling book made +$1,742 in the intervals where the timed book held nothing, with ES −$99.

- **The signal fires after the move, and shorting after it pays.** The signal turns short when IV has just jumped above the GARCH forecast. In the 143 intervals just before the signal turned short, the rolling book lost $2,879. The timed book was still holding a position, after an earlier signal, in 58 of them and lost $1,065 there. In those 58 the rolling book lost $1,780. In the intervals that followed a short signal, both books made money: in the 463 where the timed book held a position, the rolling book made +$506 and the timed book +$1,034. In the 335 intervals where the timed book was still holding after the signal had ended, both lost (−$937 and −$716). But that is the same jump effect: the 58 jumps above all fall here, and the other 277 intervals made money (+$843 and +$350). So the signal's short days were not bad days to be short. The losses come from being short when IV jumps, and the signal cannot see a jump in advance. The timed book's weakness is that it sits out the calm stretches. Those pay the rolling book more than the jumps cost it. The pattern is the same at 2x.

- **The 25-delta strike still matters.** The ATM version (`always_atm`) earns more before costs (+$3,328) but pays $2,967 in spreads against $453, and loses at both spreads. This repeats the `2026-10-17-otm-put` finding: ATM closes, especially deep in the money, are expensive in the spread model.

- **2022 is the surprise.** The timed book lost $1,039 in 2022; the rolling book broke even. In 2022 the rolling book lost $1,143 in the intervals where the timed book held a position (the timed book lost $1,102 there) and made $1,117 in the rest. So it lost as much as the timed book while the timed book held a position, and the premium it collected between those stretches is what saved it.
- **The tail is spread over the stress years.** The rolling book's worst 5% of days fall mostly in 2022 (30 days), 2020 (22), then 2024, 2018 and 2023. 60 of its 129 tail days are also in the timed book's tail. The worst day is −$677 on 2024-08-02 (a stop). The losing years are 2018 (February and December), 2019 and 2024.

4 Verdict

Failed on the pre-registered bar, on the tail guard only (ES 1.68x the reference at 1x and 1.69x at 2x). The P&L hypothesis held clearly. Not promoted. The ES limit was set before the run and kept by the PM, so it is not changed now.

What this means for the desk: the most profitable book so far, and the only one positive at 2x, is the simplest. It is always short a 25-delta put, hedged with a 10-share band. Whether its tail is acceptable is a risk decision for the PM, and it is not answered by re-testing this card. Possible follow-ups, each needing its own card (all in-sample, each adding trials):

- **A risk-adjusted bar:** compare the two books on a pre-registered risk-adjusted measure, e.g. P&L per unit of ES. ES over the timed book's holding intervals does not rescue the rolling book: there it is the riskier of the two.
- **A less biased forecast:** the timed book loses because it is flat in calm stretches. `2026-10-07-garch-bias-correction` found that the GARCH forecast runs above realized vol (by 3.1 points on average). A forecast that tracks calm regimes better is the signal-side test. Earlier attempts failed: the trailing bias correction (`2026-10-07-garch-bias-correction`) and HAR on squared daily returns (`2026-10-05-har-vs-garch`, `2026-10-16-har-signal`), which ran even higher in calm years. An untried option is HAR on intraday realized variance. R2 has no intraday SPY spot, so this would use the replay's put-call-parity spot. Skipping the signal's short days is not supported: shorting after a short signal made money in both books. A stress filter in the VIX-gate style is also less promising here: 60 of the rolling book's 129 tail days are also in the timed book's tail, where the VIX gate already failed (`2026-10-16-vix-gate-backtest`).
- **Further out of the money (15-20 delta) on the rolling book.**

5 Caveats

- **In-sample:** the idea comes from the desk's review of the baseline signal and from the archived v1 results, which cover 2015-2021 (inside this window). The proxy test in `2026-10-07-garch-bias-correction` warned that shorting more often adds tail losses, and the tail result here agrees.
- **Spread model:** fitted on about two weeks of calm 2026 live quotes. The OTM-vs-ATM cost gap drives part of the result, and real paper-account fills should be checked against it.
- **Exposure:** the rolling book is short on almost every day, so it depends on the volatility risk premium staying positive. Over 2014-2024 it was positive on about 70% of days. A long stretch without it, or a bigger spike than 2020, would hit this book harder than a timed one.
- **One contract**, with a $1 order minimum. Month-end R2 days are missing (89 skipped times). `always25` also logged 9 unfilled orders, 10 days without a delta and 10 without a mark, all in 2016 and 2019 (`put25` and `always_atm` have none). Stops and profit-takes rest on 10 and 34 events.

6 Trials

4 trials for this experiment (2 variants at 1x and 2x, one run of `experiment.py` at commit 4e3e599), plus the baseline's 2. The reference was reused from `2026-10-17-otm-put` (2 trials, not rerun). After the run, the always-short lines in the exposure chart were given distinct styles and the figures rebuilt with `--report-only`, adding no trials. The pass bar was not changed. The 2026-10-18 correction re-analysed the registered runs only and added no trials.